Some additional project guidelines

Quantitative Reasoning II · Final Project

I noticed a lot of trends in the project ideas. Instead of sharing individual feedback, I've codified the most significant pieces of guidance in one document. The next step in your project is to select a topic (if you haven't already) and hone your research question. After that, begin your data analysis. Initial drafts of your project are due to Populi by 11:59 PM on Sunday November 1, 2026. Please utilize my assistance via email, office hours, or meeting as you run into questions or issues!

Please review this whole document and consider the guidelines. If you're not sure whether you understand or are on the right track with your question or data—please let me know.

Question format

Research questions for this project should be clear, specific, and of a reasonable scope. If a single model or statistic answers your question, it is too narrow. If it requires a dissertation-length investigation, it is too broad.

About half of the submitted questions had the format "is there a relationship between X and Y." To answer this you only need a correlation. I'd like your questions to go beyond "yes or no" and ask to what extent, in what way, or under what conditions are X and Y related. Here are a few examples of question formats that hit the right scope:

1. **How big the gap is between two groups:** *how large is the difference in Y between A and B, among units that are otherwise alike?* How large is the difference in fuel economy between hybrid and gasoline cars of similar size and engine power?

2. **How much one thing moves with another, once the obvious alternative is accounted for:** *how much does Y change with X, after accounting for Z?* How much does a country's life expectancy rise with health spending per person, after accounting for income per person?

3. **Whether a relationship is the same for everyone:** *how does the relationship between X and Y differ by W?* How does the relationship between temperature and household electricity use differ between summer and winter?

Data

Submissions varied a lot on data. Some of you found good datasets, opened them, and got oriented. Others described a webpage or data a model pulled for you. A few guidelines:

Data sources: The quickest way to get moving is a well-documented dataset from a reputable source, downloadable as a `.csv`. You are welcome to collect data in more advanced ways, including using an LLM to write code that scrapes a website and builds a dataset of your own. Do not ask a model to pull the data for you in the chat. That is a recipe for hallucinated and incomplete data.

Combining datasets: Some research questions need more than one dataset, possibly from different sources. Don't be afraid to do this! Think carefully about the join: you may need to define new variables, or group one dataset, before you combine.

Rows versus observations: Check what one row represents before you start. If a row is something like "percent of adults who did X in 2021," there's a good chance that your data is not granular

enough. The number of rows in a file is also not necessarily the number of observations your analysis needs. Suppose a file has 24,000 rows, and each row is one country, in one year, for one age group: 100 countries across 20 years with 12 age groups. If your question is about countries and years, you have 2,000 observations (group the 24,000 rows by country-year) rather than 24,000. The count that matters is what survives your filters and your merge.

How much data to use: Don't cut your data down for convenience: Python will process millions of rows quickly, so drop rows only when your question or dataset gives you a reason to.

Missing values: For each variable you plan to use, work out from the file what share of rows carry a value, rather than taking that figure from the documentation. If you drop rows with missing values for key variables, report how many observations are dropped. If there are clear trends in what rows are missing values, consider how this changes your sample. Check how missing values are stored, too. Many files use a code such as `-1`, `.`, `\N` instead of leaving a cell blank, which, without proper instruction, will read as real data, throwing off your analysis or causing errors in your code.

Dependent versus independent variables: Some submissions seem to be mixing these up. The dependent variable is your Y: the quantity the model predicts, the outcome you are trying to explain. The independent variables are your Xs: the characteristics or quantities you use to explain variation in Y. In a question about how genre affects a song's popularity, popularity is the dependent variable. Characteristics like age, genre, location, and sex are rarely dependent variables.

Evaluating a data source

A dataset can be well defined and still have limitations for your particular question. Before you commit to a source, understand where the data came from. What was it built for? How do observations end up in the dataset? How are variables measured? I expect you to be able to articulate the virtues and limits of your data and the effect of these things on your analysis. If you are having trouble with this, talk to me!

Language models and data

Because this isn't a coding class, you'll use a model to write your code. You can also use a model to help you improve or evaluate your analysis, or to locate potential data sources. Do not ask the model to give you data. It will make things up. Some guidelines for LLM usage in building your analysis:

- **Give specific instructions:** Specify file name, columns, filters, and judgments. The model will fill in the gaps if you are not specific, and may make inappropriate judgments.
- **Make the model audit its own work:** Use new instances of your LLM to audit your code. Imagine you have two employees. Employee A performs a task for you. You then hand the instructions and goal you gave Employee A to Employee B and ask it to check Employee A's work. The same can be performed from the lens of critique—use it to find errors or design flaws. Models catch far more when asked to criticize than when asked to produce.